CGO study: Claude, graphify and Obsidian for marola

Study behind MIP-0089. Written 2026-10-12 for Hoffmann. Every number below names the command that produced it; `(v)` marks a claim checked against a source fetched while writing, `⚠` one that was not.

1. The answer in five lines

1. “Grafiphyy” is **graphify** (PyPI `graphifyy`, Apache-2.0), the tool MIP-0076 pinned. No other tool by that name was found (v, PyPI and GitHub, 2026-10-12).
2. graphify already writes an **Obsidian vault**; marola only needs to run it in MIP-0084’s CI job and add the tags graphify leaves out: repo, MIP, status, area.
3. A code-only graph does **not** show how ideas match across repos: 16 of 641 communities span two or more repos. The idea links come from MIP-0085’s doc↔code and decisions indexes.
4. **Order**: MIP-0084 first (MIP-0085 is declared blocked by it), then CGO step 1 beside MIP-0085, then CGO step 2 once MIP-0085’s indexes exist.
5. A **marola-built viewer** costs about ten PRs (≈ $60 of agent time, ⚠ estimate) plus review, and $0 to host. Park it until people are seen using the Obsidian vault.

2. What CGO means here

Letter	Tool	Role	Already in marola?
C	Claude Code, Claude Projects	asks the graph (`route`, the routing digest)	yes: MIP-0084 vendors `route` into every repo
G	graphify 0.9.86	builds the graph from code and Markdown, offline	yes: pinned by MIP-0076, run in CI by MIP-0084
O	Obsidian	shows the graph to a person as linked notes, tags and a graph view	no

MIP-0084 and MIP-0085 are the two Accepted MIPs about the graph (v, `docs/MIPs/README.md` on `main` at `6d0a186`): MIP-0084 makes the graph reachable by every agent with no setup; MIP-0085 adds the indexes (doc↔code, impact, decisions, meaning) to the same bundle.

3. The spike (2026-10-12)

Setup. graphify 0.9.86 in a scratch venv (`uv pip install graphifyy`), every call under `unshare -rn env -i HOME=<scratch> PATH=<venv>/bin:/usr/bin:/bin OLLAMA_BASE_URL=http://127.0.0.1:9`: no network, no keys, so no model call is possible. Repos at their `main` tips of that morning. marola-research, marola-skills and marola-agents were not in this checkout and are not counted.

Table 1: one graph per repo (`graphify update . --no-cluster`)

Repo	Commit	Nodes	Edges	Code nodes	Doc nodes	Time
marola (umbrella)	6d0a186	2,380	2,855	223	2,088	3.3 s
marola-app	8d3af82	2,383	5,511	2,171	177	2.2 s
marola-site	239fbea	8,257	20,908	7,288	939	13.1 s
marola-corpus	d39b0e1	60	53	8	52	0.7 s
marola-ml	91ab55d	307	560	150	93	1.1 s
marola-oods	9f3809e	35	36	9	26	0.6 s
marola-devkit	0749822	870	1,745	579	222	1.4 s

6,949 of marola-site's nodes are the vendored `mapbox-gl-csp.js`.

Table 2: merged, clustered and exported

Step	Command	Result
merge	`graphify global add out/<repo>/graph.json --as <repo>` ×7	14,200 nodes, each with a `repo` field; 19 MB JSON
cluster	`graphify cluster-only merged --no-label --no-viz`	30,979 edges, 641 communities, 15 s, 0 tokens
vault	`graphify export obsidian --dir vault`	14,841 notes, `graph.canvas` 3.5 MB, 64 MB in all, 6 s
browser view	`graphify export html --node-limit 5000`	`graph.html` aggregated to 641 community nodes, 0.6 MB, 1 s
pruned	drop `site/static/vendor/` and vendored `.claude/skills/`	6,398 nodes, 11,253 edges

What the merged graph shows, and what it does not

- **Cross-repo edges are few and mostly unresolved.** 803 of 30,979 edges join nodes of different repos; most point at nodes with no repo (an import of an external name). The largest real pairs are marola-devkit→marola (47) and marola-app→marola (40), mostly same-name script matches.
- **Communities stay inside repos.** 16 of 641 span more than one repo.
- **The vault loses the repo.** Each note's tags are file type, edge confidence and community (`graphify/export.py`, `to_obsidian`: `node_tags = [ftype_tag, conf_tag, comm_tag]`); the `repo` field from the merge is not written. A note on `Swimability` reads:

```
---
source_file: "core/src/main/scala/marola/scoring/Swimability.scala"
type: "code"
community: "Community 26"
tags:
  - graphify/code
  - graphify/EXTRACTED
  - community/Community_26
---
```

Reading. graphify gives a navigable code map for free, and Obsidian can open it today. It does not by itself answer "how do ideas in marola-corpus relate to marola-oods": that needs the doc↔code links MIP-0085 §5.4 measured (2,126 exact backtick matches in 123 of 131 umbrella docs) turned into tags, which is CGO step 2.

4. MIP-0084, MIP-0085 and CGO in perspective

	MIP-0084	MIP-0085	CGO (MIP-0089)
Audience	every agent session	agents	people, then agents via tags
Status	Accepted 2026-10-10, 12 task issues filed	Accepted 2026-10-10, 11 task issues filed	Draft
Builds	`route`, the `routing-bundle`, a SessionStart hook, the vendored skill in six repos	doc↔code, impact, decisions, a measured semantic spike, a 30-question bench	a vault ref with tags, `knowledge-graph.yml`, `kg-enroll`
Blocked by	MIP-0076 (Implemented)	MIP-0084	MIP-0084 task 5 (step 1); MIP-0085 tasks 5–6 (step 2)
Effort	L	L	M
Value alone	high: zero sessions reach the graph today	none until MIP-0084 ships	a code map only; ideas need MIP-0085

Which goes first: MIP-0084. Three reasons, each checkable:

1. MIP-0085's header says `Blocked by: 0084`, and every row of `MIP-0085.tasks.md` names "MIP-0084 implemented" as its precondition.
2. CGO's vault is one more output of MIP-0084's bundle job (§5.3 there); without the job there is nowhere to build it.
3. MIP-0084 fixes a measured zero: no session outside its builders called the graph (MIP-0084 §2). MIP-0085 and CGO improve answers nobody receives yet.

Then, in parallel: MIP-0085's bench and indexes (agent quality) and CGO step 1 (a vault with `repo/` and `kind/` tags; a few dozen lines in a job that exists). **Last:** CGO step 2, which reads MIP-0085's `doc_code.json` and `DECISIONS.md`. The viewer waits for evidence.

5. What a marola-built visualization layer would cost

Calibration: MIP-0076 recorded about $151 of `Cost:` trailers over 24 PRs, about $6.3 per PR, almost all `cost-split --estimate-commit` diff-size estimates (v, MIP-0076 header). Person time is ⚠ an estimate, not measured.

Tier	What it is	PRs	Agent cost (≈ $6.3/PR)	Person review	Hosting	Ongoing
A: CGO	vault + tags in the bundle, `graph.html` beside it, reusable workflow, `kg-enroll`	5	≈ $30	≈ 4–6 h	$0 (a git ref)	one bump PR per graphify update
B: marola atlas	a docs.marola.dev page: vendored sigma.js + graphology (MIT), pruned graph JSON, filters by repo/MIP/area, search, GitHub deep links	+10	≈ +$60	≈ 2–3 days	$0 (GitHub Pages)	breaks when graphify's JSON changes; a monthly fix PR
C: semantic atlas	B plus graphify's LLM pass so docs, PDFs and the corpus get meaning edges	+5	≈ +$30	≈ 1 day	$0	LLM tokens, below

The LLM pass in tier C. MIP-0076 measured 0.6–0.7M input tokens per full pass. At 0.65M input tokens, run daily (19.5M a month), on the Claude API's prices as cached on 2026-10-06 (v, the `claude-api` skill's model table), input only, output not measured (⚠):

Model	Input $/1M	Per pass	Per month (daily)
Claude Haiku 5.5	$0.10	$0.07	≈ $2
Claude Sonnet 5.5	$2.00	$1.30	≈ $39
Claude Opus 5.5	$4.00	$2.60	≈ $78

A paid model call in CI is a paid resource: it needs a person's go-ahead first (umbrella `AGENTS.md`, cost rule). A local Ollama model costs $0 in money and needs a GPU runner.

Bought instead of built. Obsidian: free, Sync $4/user/month and Publish $8/site/month billed annually, commercial licence $50/user/year and optional (v, obsidian.md/pricing, 2026-10-12). Graphify Cloud: 14-day trial, no published price (v, its README, 2026-10-12).

Recommendation. Tier A now; B only if four weeks of tier A show at least three people opening the vault (MIP-0089 §7); C not at all while the semantic spike in MIP-0085 §5.7 has not shown that meaning search beats BM25 on marola's own questions.

6. Adding other repositories "in one click"

- **Inside the umbrella** (marola-corpus, marola-oods, marola-research, marola-skills, marola-agents and the rest): nothing to do. The bundle job graphs every submodule and tags each note `repo/<name>`.

- **Outside** (ww3-gpu, the awesome lists, a partner's repo): `just kg-enroll <owner/repo>` opens two PRs, a ten-line caller of the devkit's `knowledge-graph.yml` in the target and a line in the umbrella's `knowledge-sources.yml`. The target's own CI builds its graph offline and pushes it to its `refs/marola/graph`; the umbrella's job fetches it.
- **Why a PR and not a button.** The graph steers agents and readers; MIP-0084 §8 names a rewritten bundle as the way to mislead them. A person merging a reviewed file is the click.

7. Tags: the index scheme

Tag	Meaning	Source	Step
`repo/<name>`	which repo the note is from	`graphify global add --as`	1
`kind/code`, `kind/document`, `kind/rationale`	node type	graphify's `file_type`	1
`mip/MIP-NNNN`	the MIP that names this file or symbol	MIP-0085 `doc_code.json`	2
`status/draft` … `status/implemented`	that MIP's status	the MIP's Status row	2
`area/<a>`	GitHub label of the MIP's issues	`labels.yml` `area/*`	2
`decision`	a row of `DECISIONS.md`	MIP-0085 `decisions.py`	2

graphify's own `graphify/*` and `community/*` tags stay. An ambiguous doc↔code link (a third of them, MIP-0085 §2) is listed in the note, not tagged.

8. Risks

- Nobody opens the vault; the step-1 cost is small, the viewer stays parked.
- Code-only edges are not ideas; without step 2 the vault is a code map.
- graphify moved from 0.9.66 to 0.9.86 in ten days; the export is pinned with graphify.
- Obsidian is closed source; the vault is plain Markdown and `graph.html` needs no Obsidian.
- marola-research has no content yet (REPOS.md: "Science", no contract).

9. Reproducing the spike

```
        uv venv gv && uv pip install --python gv/bin/python graphifyy==0.9.86
        G="env -i HOME=$PWD/home PATH=$PWD/gv/bin:/usr/bin:/bin GRAPHIFY_NO_AUTO_REFRESH=1 \
           OLLAMA_BASE_URL=http://127.0.0.1:9 unshare -rn graphify"
        for r in marola marola-app marola-site marola-corpus marola-ml marola-oods marola-
devkit; do
          (cd "../$r" && GRAPHIFY_OUT="$OLDPWD/out/$r" $G update . --no-cluster)
          $G global add "out/$r/graph.json" --as "$r"
        done
        mkdir merged && cp "$($G global path)" merged/graph.json
        $G cluster-only merged --graph merged/graph.json --no-label --no-viz
        $G export obsidian --graph merged/graphify-out/graph.json --dir vault
        $G export html --graph merged/graphify-out/graph.json --node-limit 5000
```

Sources (fetched 2026-10-12)

- PyPI: graphifyy and its JSON API (version, licence, upload date).
- Graphify-Labs/graphify README (Cloud, LLM use).
- Obsidian pricing and licence.
- npm registry: sigma, graphology, cytoscape.
- MIP-0076, MIP-0084, MIP-0085 on `main` at `6d0a186`.